

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT INITIAL PROPOSAL

[Edu Path (Academic & Career Advisor)]

Project Team

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Project Proposal

Project Title: AI powered Academic and Career Decision Support System

Description:

The proposed project is about AI based personalized academic and university recommendation system which will help those students that have just completed their intermediate education and planning for higher education. During this process many students face difficulty while deciding which academic field is good for them, which degree program is suitable for them and which university would be a good option for them. Students have different educational backgrounds of subjects, grades, interests, skills, career goals, financial limitations, and location preferences, but they often do not have a proper way to connect with all these factors when making an academic decision. The proposed system aims to solve this problem by analyzing a student's overall profile and providing personalized recommendations at three levels: a suitable academic field, suitable academic programs, and a suitable universities.

The system will start by collecting information from the students. This includes their education, what subjects they studied, their grades, what they're good at what they like, what skills they have what they want to do in the future, where they want to live, how much money they have and any other preferences. Students can also explain their interests and goals in their own words. The system will use natural language processing to understand this information and first decide which field of study fits best. For example, a student who likes programming, mathematics, artificial intelligence, software development and solving problems might be matched with the field of computing or computer science. Another student who is interested in business, management, finance and entrepreneurship might be matched with the business field. This part helps the system understand the student's direction.

Once the right field is found, the system will look closely to find the best programs. A field can have different programs so the system will look at the student's subjects, interests, skills and future goals to find the best match. For example, if the field is computing, the system might suggest intelligence, computer science, software engineering, data science, or cyber security depending on what the student is like. The programs will be ranked based on how they fit the student. So, the system will not just tell the student which field is right, but also which specific programs fit them best.

The next step is to recommend universities. The system will use the programs that have been recommended and the student's preferences to find suitable universities. It will

consider factors such as whether the university offers the recommended program, whether the student meets the admission requirements, the type of university, its location, the student's budget, scholarship opportunities, distance, and other preferences provided by the student. The system will have a list of recognized universities along with information about the programs they offer. First we remove universities that do not fit the student requirements. Then we rank the remaining universities with a decision-making method such, as TOPSIS or MCDM. This method lets the system consider factors when ranking universities instead of relying only on the university's name or reputation. For example two students might get the program recommendation but at the same time they might receive different university recommendations because they may have different budgets, preferred locations, eligibility criteria, and other personal requirements.

The entire system operates in three steps: first the system finds the desired field of study; second, it recommends and ranks the programs; third it locates, verifies and ranks the best universities that provide those programs and satisfy each student's needs.